DARSHIL SHAH

Embedded Software Engineer | Real-Time Systems | Industrial IoT

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Professional Summary

Embedded Software Engineer with 6+ years of experience in automotive, agricultural, and industrial IoT systems.
Expertise in Embedded C/C++, Python, C#, FreeRTOS, CAN/J1939, and MATLAB/Simulink model-based development.
Strong background in system integration, debugging, and CI/CD automation for scalable embedded platforms.
Focused on delivering reliable, safety-critical real-time control systems with measurable impact.

Core Expertise

Real-time embedded firmware development (Embedded C/C++, FreeRTOS)
CAN / J1939 communication and diagnostics
Control system software development and integration
Firmware + GUI integration using C#/.NET
CI/CD automation (GitLab CI/CD, Jenkins, Python)
Embedded debugging (CANalyzer, Wireshark, JTAG, logic analyzer)
Workflow Automation

Technical Skills

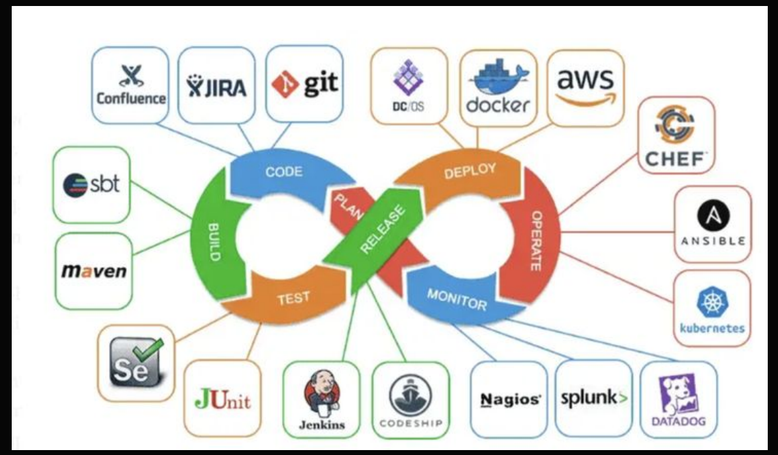
Languages: C, C++, Embedded C, Python, C#, MATLAB, Bash

MCUs/Platforms: STM32, TI Tiva, ARM Cortex-M, Renesas, Apollo3, Raspberry Pi

Protocols: CAN, J1939, SPI, I2C, UART, Ethernet, UDP, BLE

RTOS/OS: FreeRTOS, Embedded Linux, Custom RTOS

Tools: Git, GitLab, Jenkins, Docker, SVN, CANalyzer, Wireshark



Experience | CNH Industrial (Sprayers)

Developed suspension and axle control software improving vehicle stability by 30%.
Automated build and deployment using Python + GitLab CI/CD, reducing release time by 40%.
Led SVN → Git migration improving collaboration and reducing merge conflicts by 50%.
Integrated embedded firmware with C# GUI tools, reducing latency by 25%.
Implemented MATLAB/Simulink code generation, reducing integration issues by 40%.
Resolved 100+ issues in Polarion/SharePoint contributing to 30% reduction in field failures.

Internship Experience

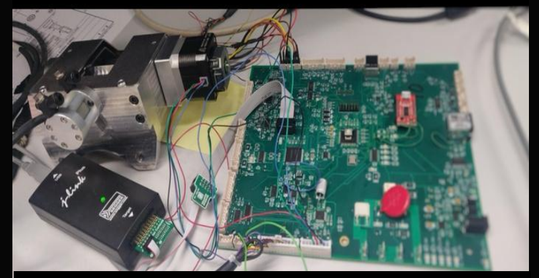
The Coats Company (Hennessy Industries) – Embedded Software Intern

- Built SPI interface between FreeRTOS and AMT203 encoder
- Implemented stepper motor stall detection improving safety diagnostics
- Developed TI ↔ Renesas inter-MCU communication protocol

Sloan Valve Company – R&D Embedded Software Intern

- Developed BLE-based OTA firmware upgrade workflow
- Performed PCB bring-up, validation, and failure analysis

OTA firmware update



Project Spotlight | Custom Preemptive RTOS

Goal: Build lightweight RTOS for deterministic real-time execution on ARM Cortex-M4.

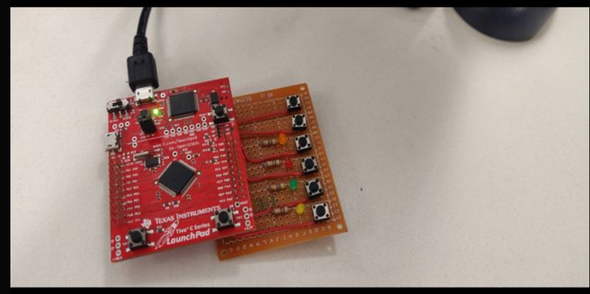
Designed scheduler and dispatcher using SysTick and PendSV.

Implemented task APIs: create/destroy/sleep/yield/priority control.

Developed semaphores and mutex synchronization primitives.

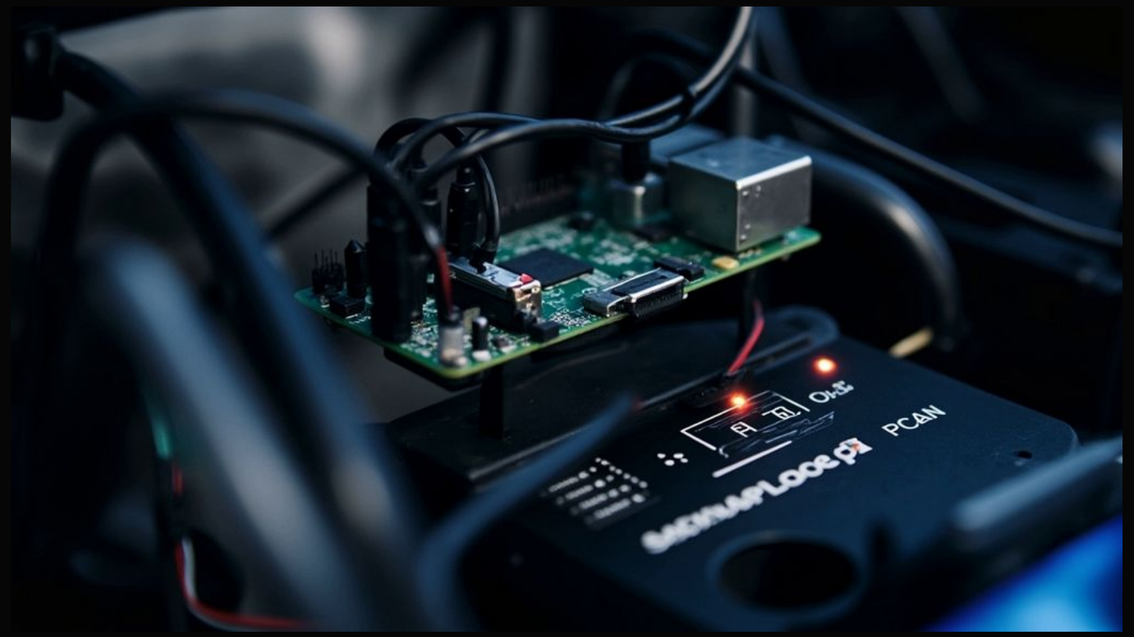
Built UART CLI for runtime debugging and monitoring.

Result: <10μs context switch latency and stable multitasking with 6 tasks.



Project Spotlight | CAN J1939 Fault Prediction

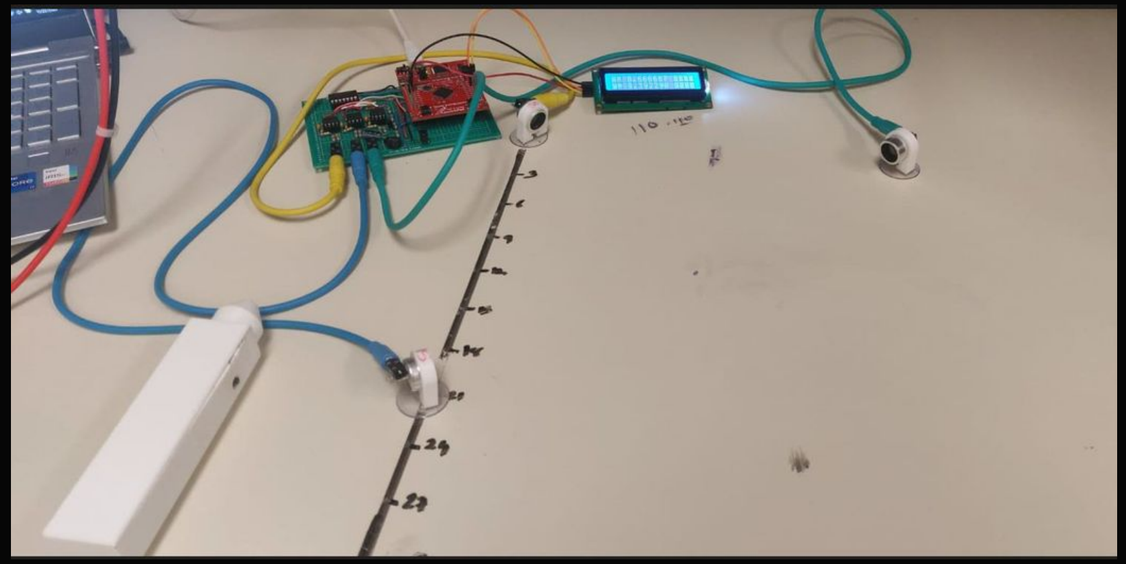
Goal: Predict DM1 fault behavior using CAN datasets.
Built preprocessing pipeline for CAN diagnostic frames.
Developed predictive fault classification model using Python + ML.
Result: Achieved 99% prediction accuracy for fault classification.



Project Spotlight | SONAR Object Localization

Goal: Real-time object localization using ultrasonic sensors.
Implemented time-of-flight processing and triangulation logic.
Built UART-based CLI for calibration, diagnostics, and parameter tuning.
Result: Achieved sub-centimeter localization accuracy (±5mm).

https://drive.google.com/file/d/1HSdQGvz_kPaBjFFxB7GusVK1dN65bjrT/view?usp=sharing



Project Spotlight | Workflow Automation (n8n + Gemini)

Goal: Automate AI-based email drafting workflow.

Built n8n automation pipeline integrated with Google Sheets + Gemini API.

Implemented sequential processing and status-based skipping.

Result: Improved productivity through scalable automation execution.

Additional Projects

Peer-to-Peer CSMA Protocol – Designed low-latency embedded communication protocol.
CNC Network Interface (UDP → SPI Bridge) – Enabled real-time CNC control.
OTA Firmware Upgrade (Apollo3 Blue SoC) – Built remote firmware update mechanism.
Scalable IoT Data Ingestion System (HTTP-based) – Embedded-to-cloud ingestion pipeline.

Education & Contact

M.S. Electrical & Computer Engineering – University of Texas at Arlington (GPA: 3.9)
B.E. Instrumentation & Control Engineering – L.D. College of Engineering (GPA: 3.4)

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